SUREN NETWORK
PLAY WITH TECHNOLOGIES

# SNCNA

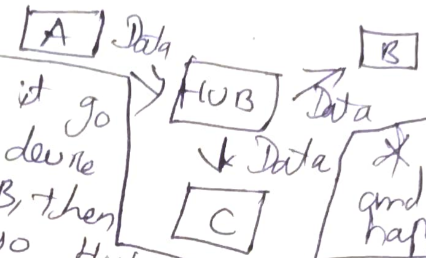
SUREN NETWORK CERTIFIED NETWORK ASSOCIATE

## DEVICE TEST

**1, Explain behaviour of hub device**

1) Hub is a device transfer network to multiple device
2) Layer 1 (physical layer) transfer data as Binary ⇌ Electrical
3) It is Broadcast device

A to B



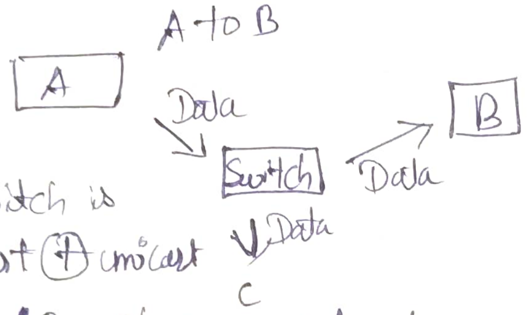

4) If A sent to B a data, first it go to Hub, Hub will broadcast to all device than Hub will receive data from B, then also Hub will broadcast

* Network trafficking and collision can happen, it is said to be Half duplex.

**2, Explain behaviour of switch device**

1) Switch is a device that transfer the network and data to multiple devices
2) It is also a broadcast device, 3) It is layer 2 device (Datalink layer)
↓
MAC

A to B



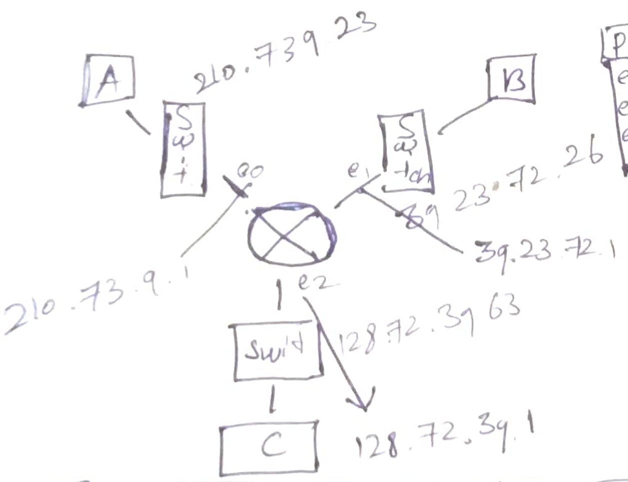

A Sent a data to B, first it reach the Switch then Switch will broadcast to all device, then B receive it Sent data to switch, then Switch will unicast to A rather than broadcast (with help of MAC Table)

So switch is broadcast + unicast

∴ ARP plays a vital role to Send MAC and receive MAC

**3, Explain behaviour of router device**

A router sends is device that only allow unicast, it also transfer data with different network.




Routing table

| P.I | N.ID / Prefix |
|---|---|
| e0 | 210.73.9.0/24 |
| e1 | 39.0.0.0/8 |
| e2 | 128.72.0.0/16 |

* A Send a data to B, first it reach the Swith
* If it is same network, it won't go to router, if it is diff network it goes to router

* Now router note the ip as 210.73.9.1 as first IP for easy understanding, now it look Interface, Network ID, prefix in routing table. (look 39.0.0.0/8 e1)
* Now router send the data to dest ip 39.23.72.26 because the ips N.ID, prefix, interface in routing table.
* then it goes to Switch, Switch will broadcast then reach dest ip. then B send network to A, reach Switch diff network, go to router, router check routing table (210.73.9.23) in routing table then it goes to A.

that how data transfer in router

* Routing table play imp role
* It is a unicast device